

# Crowded Carry: Crash Risk in Foreign Exchange and Crypto Carry Trades

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## Abstract

Do carry trades crash when they are crowded? We study G10 currency carry (1999–2026) and crypto funding-rate carry on Binance perpetual futures (2020–2026). FX carry earns a net Sharpe ratio of 0.37 and loses when volatility rises. Crowding measured from CFTC speculator positions does not predict FX carry returns or crashes in portfolio, panel, or tail tests. Crypto carry collects funding income, has no directional exposure to Bitcoin, and crashes when single coins with extreme funding collapse. The two strategies are nearly uncorrelated. Crowding tells investors where carry risk sits, not when it will materialise.

**JEL classification:** F31, G12, G15, G23

**Keywords:** carry trade, crash risk, crowding, CFTC positioning, cryptocurrency, perpetual futures, funding rates

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# 1 Introduction

Uncovered interest parity (UIP) says that a currency with a high interest rate should depreciate by exactly the interest differential, so that borrowing in a low-rate currency and lending in a high-rate one earns nothing on average. Since [Hansen and Hodrick \(1980\)](#) and [Fama \(1984\)](#), a large literature has shown the opposite: high-yield currencies tend to appreciate, or depreciate by less than the rate gap, and the carry trade earns a sizeable premium. The leading explanation is that this premium compensates for crash risk. Carry trades go “up the stairs and down the elevator”: they earn small steady gains most of the time and lose a large amount in a few weeks when global risk appetite falls ([Brunnermeier et al., 2008](#); [Menkhoff et al., 2012](#); [Farhi and Gabaix, 2016](#)).

[Brunnermeier et al. \(2008\)](#) add a second ingredient. Crashes are worse when speculators hold large positions in the trade, because a funding shock forces them to unwind together. Crowding is therefore a natural candidate for predicting *when* carry will crash. This paper asks a simple question: **do carry trades crash when they are crowded?**

We study the question in two markets that share the economics of carry but differ in who trades. The first is the classic G10 currency market, where carry is the gap between short-term interest rates and crowding can be measured from the Commodity Futures Trading Commission (CFTC) report on speculators’ futures positions. The second is the market for crypto perpetual futures. A perpetual future has no expiry; instead, longs and shorts exchange a *funding rate* every eight hours that keeps the futures price close to spot. When leveraged traders crowd into long positions, longs pay shorts. The funding rate is therefore both the crypto analogue of an interest-rate differential and a direct measure of crowded leveraged demand.

We build two pre-specified, dollar-neutral strategies. Strategy A is a monthly G10 carry strategy that buys the two currencies with the highest three-month interest rates and sells the two with the lowest, and halves its positions when CFTC positioning shows that its longs are crowded long and its shorts crowded short. Strategy B is a weekly crypto carry strategy that shorts the third of coins with the highest seven-day funding and buys the third with the lowest. All parameters, the in-sample/out-of-sample split, and transaction costs were fixed before any backtest was run.

We find four main results. First, G10 carry earns a modest premium after costs (Sharpe ratio 0.37 over 1999–2026) and closely tracks the academic carry factor of [Lustig et al. \(2011\)](#), with a correlation of 0.82. Its losses line up with volatility shocks: in a regression on the dollar factor and equity-volatility innovations, the volatility beta is  $-1.55$  ( $t = -3.7$ ) and the alpha is not significant. This is the standard risk-based picture of carry.

Second, CFTC crowding does not predict FX carry. Carry earns 0.02% in the month after a crowded reading against 0.30% otherwise, but the difference is not significant ( $t = -0.57$ ), and it shrinks once we drop the years in which the crowding signal is unavailable. Stronger tests give the same answer. In a panel of 2,149 currency-months, a currency’s own crowding z-score does not

predict its next-month excess return ( $t = 0.45$ ); the portfolio crowding level does not predict carry returns at horizons of one to twelve months; and crowded months are not more likely to be crash months. The 2008 crash started from a *negative* crowding reading. A filter that halves exposure when crowded raises the out-of-sample Sharpe ratio from 0.27 to 0.32 under every parameter setting we try, but this gain is not statistically significant ( $t = 1.0$ ).

Third, crypto funding carry collects real funding income of about 8.5% a year, but price moves decide its performance. Net of costs it earns 4.1% a year with 30% volatility (Sharpe 0.14), losing in 2020–2023 and earning a Sharpe of 0.81 since 2024. The strategy has no directional Bitcoin exposure: the beta is precisely estimated near zero (daily  $-0.02$ , 95% confidence interval  $[-0.06, 0.03]$ ) at daily, weekly, and monthly horizons, with no downside asymmetry, and its crashes are coin-specific. They arise in the mirror image of FX: because the strategy buys coins with the most negative funding, it ends up long whatever traders short hardest. In May 2022 that meant being long Terra (LUNA) into its collapse, which cost 27% in a single week.

Fourth, FX and crypto carry are almost uncorrelated (0.11). A 50/50 mix scaled to equal volatility has lower volatility and a smaller drawdown than either strategy alone.

Our answer to the main question therefore differs across markets. In FX, the answer is no: carry crashes are compensation for exposure to global volatility, and CFTC positioning adds no predictive power. In crypto, crowding measured by funding tells an investor *where* the danger is, in the coins with extreme funding on either side, but not *when* it will hit.

**Related literature.** Our paper connects three strands of work. The first documents the carry premium and explains it with risk. [Lustig and Verdelhan \(2007\)](#) and [Lustig et al. \(2011\)](#) show that a high-minus-low carry factor prices the cross-section of currency portfolios, and [Lustig et al. \(2014\)](#) show that currency risk premia are countercyclical. [Menkhoff et al. \(2012\)](#) identify global FX volatility as the priced risk, [Burnside et al. \(2011\)](#) discuss peso problems, [Jurek \(2014\)](#) studies crash-hedged carry, [Daniel et al. \(2017\)](#) document carry drawdowns, and [Farhi and Gabaix \(2016\)](#) model rare disasters. [Koijen et al. \(2018\)](#) show that carry predicts returns in many asset classes. We replicate the standard G10 result and its link to volatility risk.

The second strand studies crowding and forced unwinding. [Brunnermeier and Pedersen \(2009\)](#) model the spiral between market liquidity and funding liquidity, [Pedersen \(2009\)](#) describes how crowded trades crash when “everyone runs for the exit,” and [Stein \(2009\)](#) discusses how crowding by sophisticated investors can destabilise prices. [Brunnermeier et al. \(2008\)](#) show that speculator positions in currency futures predict the skewness of carry returns, and [Moskowitz et al. \(2012\)](#) use the same CFTC data to study how speculators trade on trends. Other studies measure crowding without futures data: [Pojarliev and Levich \(2011\)](#) infer it from the returns of currency funds and find that carry was crowded just before the 2008 unwind, and [Sokolovski \(2016\)](#) measures it from comovement among carry currencies and hedge fund exposures and finds that high crowding doubles

the probability of an extreme carry loss.

The third strand studies cryptocurrency returns. [Liu and Tsyvinski \(2021\)](#) and [Liu et al. \(2022\)](#) document the risk factors that explain crypto returns, [He et al. \(2022\)](#) describe the pricing of perpetual futures, and [Schmeling et al. \(2023\)](#) show that crypto carry, the futures–spot basis, reflects leveraged demand from retail investors and predicts crash risk. [Christin et al. \(2023\)](#) document the high Sharpe ratio of the Bitcoin funding trade, its link to the leverage that exchanges allow, and its losses in the Terra and FTX collapses. These papers study the *time-series* carry trade, long spot and short the perpetual of the same coin.

**Recent evidence (2024–2026).** A new wave of work, much of it prompted by the August 2024 unwind of the yen carry trade, revisits both strands. On the FX side, [Aquilina et al. \(2024\)](#) show that speculative short yen positions in CFTC data had reached record highs before the unwind, and [Bank for International Settlements \(2024\)](#) document that hedge funds’ exposure to carry had risen steadily since 2022 across many strategy types. These studies describe the build-up of crowding in one episode, in the spirit of [Brunnermeier et al. \(2008\)](#), but do not test whether positioning predicts crashes systematically. [Cheema-Fox et al. \(2025\)](#) measure positioning from institutional investor flows instead of futures data and show that crises differ: some are flights to safety into the dollar, while August 2024 was a flight from carry centred on the yen, and flows around the April 2025 tariff announcement were driven more by carry than by safe-haven demand. [Kutuk and van Wijnbergen \(2025\)](#) use option-implied crash measures for dollar–lira carry and find that 46–77% of carry returns compensate for crash exposure, extending the risk-based view of [Menkhoff et al. \(2012\)](#) and [Farhi and Gabaix \(2016\)](#) to forward-looking measures. Together these papers suggest that crowding matters, but they measure it with flows, fund exposures, or option prices rather than CFTC positions, which is the gap our FX tests address.

On the crypto side, [Schmeling et al. \(2025\)](#) update the evidence of [Schmeling et al. \(2023\)](#): the basis reflects an “inconvenience yield” of holding spot, driven by trend-chasing, leveraged retail demand and by arbitrage capital that is scarce because of margin and regulatory limits. [Borri et al. \(2025\)](#) show that the Bitcoin funding trade studied by [Christin et al. \(2023\)](#) earned a Sharpe ratio above six over 2020–2025, but that its premium compressed sharply after 2024 and turned negative in 2025. [Gornall et al. \(2025\)](#) show that perpetual futures reduce extreme price dislocations and the drawdowns of arbitrage strategies relative to dated futures. All three study the time-series trade in one coin at a time, mostly Bitcoin. None studies the cross-section of coins sorted on funding, the setting that corresponds most closely to the FX carry trade, and none compares crypto carry with FX carry. Our paper fills these gaps. Table I summarises these studies, their methods, and their findings, and how each relates to our results.

**Contributions.** Relative to this literature, the paper makes four contributions.

**Table I:** Related Literature: Data, Methods, and Findings

| Study                                              | Market and data                                                | Method                                                                     | Main finding                                                                                                      | Relation to our paper                                                 |
|----------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <i>Panel A: Carry as a risk premium</i>            |                                                                |                                                                            |                                                                                                                   |                                                                       |
| Lustig et al. (2011)                               | 35 currencies, 1983–2009                                       | Currency portfolios sorted on forward discount; factor model               | A high-minus-low carry factor and the dollar factor price currency portfolios                                     | Benchmark: our Strategy A has a 0.82 correlation with their factor    |
| Menkhoff et al. (2012)                             | 48 currencies, 1983–2009                                       | Asset pricing tests with global FX volatility innovations                  | Carry loses when global FX volatility rises; volatility risk explains the premium                                 | Confirmed: volatility beta $-1.55$ ( $t = -3.7$ )                     |
| Kutuk and van Wijnbergen (2025)                    | USD–TRY, survey expectations and options                       | Option-implied crash measures; Shapley decomposition                       | Crash risk raises expected carry; 46–77% of returns compensate crash exposure                                     | Supports the risk view; options suggested as a faster crowding signal |
| <i>Panel B: Crowding and carry crashes</i>         |                                                                |                                                                            |                                                                                                                   |                                                                       |
| Brunnermeier et al. (2008)                         | 8 currencies, CFTC positions, 1986–2006                        | Predictive regressions of skewness on interest differentials and positions | Speculator positions predict negative skewness; unwinds follow funding-liquidity shocks                           | We test returns and crashes instead of skewness: no predictive power  |
| Pojarliev and Levich (2011)                        | Currency fund returns                                          | Crowding from co-movement of fund style exposures                          | Carry was crowded just before the 2008 unwind                                                                     | Uses fund returns, not futures positions                              |
| Sokolovski (2016)                                  | G10 FX, hedge funds, daily                                     | Crowding from abnormal FX correlation and fund exposures                   | High crowding doubles the probability of an extreme carry loss                                                    | Faster measure that finds an effect; contrasts with our CFTC null     |
| Aquilina et al. (2024)                             | Yen carry unwind, August 2024                                  | Event study of positions, margins, volatility                              | Record speculative short yen positions were unwound; deleveraging amplified the moves                             | Our crowding measure was high before the July 2024 loss               |
| Bank for International Settlements (2024)          | Hedge fund returns                                             | Rolling betas of fund returns on carry                                     | Hedge fund carry exposure rose steadily from 2022; leverage amplifies crowding                                    | Describes build-up; no systematic prediction test                     |
| Cheema-Fox et al. (2025)                           | Institutional flows (custody data)                             | Panel regressions of flows in crises                                       | Crises split into flights to safety and flights from carry (yen in 2024)                                          | Flow-based crowding; supports faster measures                         |
| <i>Panel C: Crypto carry and perpetual futures</i> |                                                                |                                                                            |                                                                                                                   |                                                                       |
| Christin et al. (2023)                             | BTC and 17 coins, Binance, 2020–2023                           | Time-series trade: long spot, short perpetual                              | Very high Sharpe for BTC; falls when exchange leverage is cut; losses in Terra and FTX                            | We study the cross-section of coins instead                           |
| Schmeling et al. (2023, 2025)                      | Crypto futures and spot prices                                 | Basis decomposition; predictive regressions                                | Basis averages over 10% a year, reflects leveraged retail demand and scarce arbitrage capital, predicts crashes   | Same mechanism appears in our coin-level crashes                      |
| Borri et al. (2025)                                | BTC perpetuals, 2020–2025                                      | Time-series carry returns                                                  | Sharpe above 6, but the premium compressed after 2024 and turned negative in 2025                                 | Matches our fall in funding income (12.3% to 3.1%)                    |
| Gornall et al. (2025)                              | Crypto perpetual and dated futures                             | Empirical comparison; model of constrained arbitrage                       | Perpetuals reduce dislocations and arbitrage drawdowns                                                            | Basis trade suggested as an extension                                 |
| This paper                                         | G10 FX 1999–2026 with CFTC data; 12 coins on Binance 2020–2026 | Pre-specified long-short strategies; panel, predictive, and tail tests     | CFTC crowding does not predict FX carry; crypto carry risk is coin-specific; FX and crypto carry diversify (0.11) |                                                                       |

This table summarises the studies closest to ours. “Market and data” gives the main sample; “Method” the main empirical approach; “Main finding” the headline result as reported by the authors; and “Relation to our paper” how our results compare.

1. *A comprehensive test of CFTC crowding as a predictor of carry returns and crashes.* Brunnermeier et al. (2008) show that speculator positions predict the *skewness* of carry returns. We ask whether they predict carry *returns* and *crashes*, and whether an investor could use them in real time. We test this with a pre-specified trading rule evaluated out of sample through 2026, a currency-level panel of 2,149 observations, predictive regressions at horizons of one to twelve months, and tests on the left tail of returns. All give the same answer: CFTC crowding does not predict FX carry. Read together with Pojarliev and Levich (2011), Sokolovski (2016), and Cheema-Fox et al. (2025), who find that crowding matters when it is measured from fund returns, currency co-movement, or investor flows, our evidence suggests that weekly futures positions are too slow and too partial a proxy for the crowding that drives carry crashes.
2. *A cross-sectional crypto carry strategy built on funding rates.* Existing studies of crypto carry (Christin et al., 2023; Schmeling et al., 2025; Borri et al., 2025) examine the time-series trade that holds a coin and shorts its perpetual. To our knowledge, we are the first to study the cross-sectional analogue of FX carry: long the coins with the lowest funding and short those with the highest, on a universe that keeps the coins that collapsed (Terra and FTX Token) to avoid survivorship bias. We decompose its return into funding and price components and show that funding income is real but price moves dominate.
3. *Evidence that crypto carry risk is coin-specific, not market-wide.* We show that the cross-sectional strategy has no directional Bitcoin exposure, with a beta precisely estimated near zero at every horizon and no downside asymmetry, and that its crashes come from a few coins at the extremes of the funding distribution. The worst losses arise from being long coins that traders were crowding to short (Terra and FTX Token in 2022) and short coins that retail traders were crowding to buy (Dogecoin and XRP in 2021). This links the leveraged-demand mechanism of Schmeling et al. (2025) to the cross-section of coins.
4. *A side-by-side comparison of FX and crypto carry under one design.* Using the same pre-specified framework in both markets, we show two different channels of crowding risk. In FX, crashes are driven by global volatility shocks that hit carry regardless of positioning. In crypto, the funding rate is both the carry signal and the crowding signal, and crowding determines *where* losses occur. Because the two strategies are nearly uncorrelated (0.11), combining them lowers volatility and drawdowns. All code, data, and results are public, so every table and figure can be reproduced.

The rest of the paper is organised as follows. Section 2 states the hypotheses. Section 3 describes the data, and Section 4 the strategies and methodology. Section 5 presents the FX results and Section 6 the crypto results. Section 7 studies the combined portfolio, Section 8 reports robustness checks, Section 9 discusses the results and their limitations, and Section 10 concludes.

## 2 Hypotheses

We split the main question into six research questions: (RQ1) whether G10 carry earns a premium after costs and reproduces the academic carry factor; (RQ2) whether crowding predicts weaker or more crash-prone carry; (RQ3) when FX carry loses money; (RQ4) whether crypto funding carry earns a premium and whether it comes from funding or from price moves; (RQ5) what drives crypto carry losses; and (RQ6) whether FX and crypto carry diversify each other. The corresponding hypotheses, written down before any backtest was run, are:

**H-A1.** G10 carry earns a positive expected excess return after transaction costs. The null is a zero expected return.

**H-A2.** Carry is weaker and more negatively skewed in the month after a crowded reading, so that cutting exposure when crowded raises the out-of-sample Sharpe ratio. The null is that next-month carry is the same after crowded and non-crowded months.

**H-A3.** Carry loses when volatility rises. The null is that carry returns are unrelated to volatility shocks.

**H-B1.** Shorting high-funding coins and buying low-funding coins earns a positive return after costs, mainly from the funding it collects.

**H-B2.** Because it is dollar-neutral, crypto carry has little exposure to Bitcoin, and its crashes are coin-specific rather than market-wide.

**H-D.** FX carry and crypto carry have low correlation, so a combined portfolio diversifies.

## 3 Data

All data are public and were downloaded on October 2, 2026. Table II summarises the sources.

**Foreign exchange.** Spot rates for JPY, GBP, CHF, CAD, AUD, and NZD begin in 1971 and EUR in 1999. Interest-rate histories are long for USD, EUR (Deutsche mark before 1999), GBP, CAD, AUD, and NZD, but begin only in 2002 for JPY and 1999 for CHF. Strategy A therefore starts in May 1999 and runs to August 2026 (328 months). Table III reports summary statistics. The Australian and New Zealand dollars have the highest average interest rates relative to the US dollar and the yen and Swiss franc the lowest, which is why these four currencies dominate the carry portfolio. Uncovered interest parity fails in the direction carry exploits: the high-rate currencies did not depreciate on average, and the Swiss franc appreciated by 2.3% a year, more than offsetting its low rate.

**Table II:** Data Sources

| Data                           | Source                                                                | Series and coverage                                                | Used for                        |
|--------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------|
| FX spot rates                  | FRED (Federal Reserve H.10), daily, sampled at month-end              | JPY, EUR, GBP, CHF, CAD, AUD, NZD against USD                      | Strategy A returns              |
| 3-month interbank rates        | OECD via FRED, monthly                                                | US, JP, euro area, UK, CH, CA, AU, NZ, DE                          | Strategy A signal               |
| Speculator positions           | CFTC Commitments of Traders, legacy futures-only, weekly              | CME currency futures, 1986–2026                                    | Crowding signal                 |
| Crypto prices, volume, funding | Binance public archive, USDT-margined perpetuals                      | Daily candles and 8-hourly funding, 2020-01 to 2026-08             | Strategy B                      |
| Carry factor portfolios        | <a href="#">Lustig et al. (2011)</a> , <i>Currency-Portfolios.xls</i> | Developed HML, dollar factor RX, volatility innovations, 1983–2021 | Validation and risk regressions |

This table lists the data used in the paper. FX spot rates are quoted in US dollars per unit of foreign currency. Interest rates are three-month interbank rates in percent per year. CFTC positions are non-commercial long and short futures positions and open interest. Crypto data cover twelve coins: BTC, ETH, BNB, XRP, ADA, DOGE, SOL, LTC, LINK, AVAX, LUNA, and FTT.

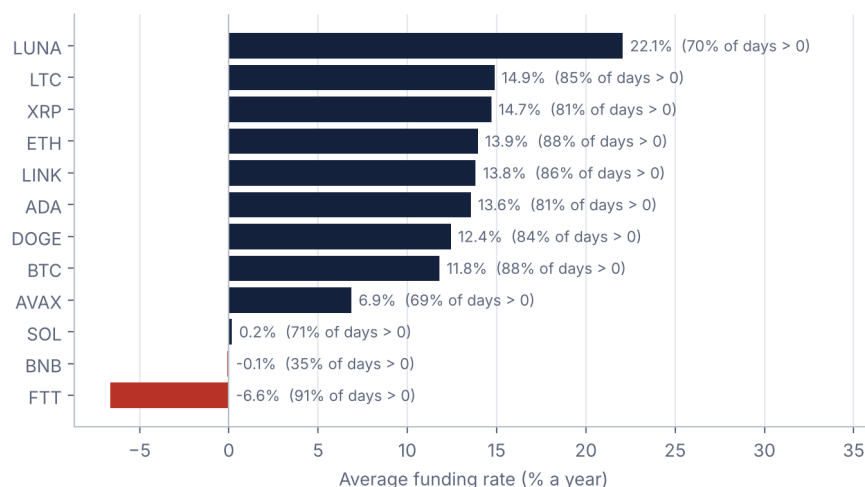
**Table III:** Summary Statistics for G10 Currencies

| Currency | Months | Rate differential (%) | Spot change (%) | Excess return (%) | Volatility (%) |
|----------|--------|-----------------------|-----------------|-------------------|----------------|
| AUD      | 327    | 1.47                  | 0.29            | 2.38              | 11.72          |
| NZD      | 327    | 1.79                  | 0.20            | 2.75              | 12.60          |
| GBP      | 321    | 0.45                  | −0.63           | 0.23              | 8.54           |
| CAD      | 327    | −0.16                 | 0.18            | 0.34              | 8.26           |
| EUR      | 327    | −0.64                 | 0.35            | 0.07              | 9.17           |
| JPY      | 291    | −1.74                 | −1.06           | −2.17             | 9.28           |
| CHF      | 324    | −1.85                 | 2.32            | 0.87              | 9.71           |

This table reports, for each currency against the US dollar, the average three-month interest-rate differential, the average log change in the spot rate (positive values mean the currency appreciated), and the mean and volatility of the monthly excess return  $rx_{k,t}$  of a long position, all annualised in percent, May 1999 to August 2026. Months is the number of monthly observations with both a spot rate and an interest rate.

**CFTC positions.** CFTC positions for CAD, CHF, EUR (Deutsche mark before 1999), GBP, and JPY are available from 1986, AUD from 1987, and NZD from 1999, with a gap in the NZD contract between 2000 and 2003. Before August 2000 the CFTC lists currency futures under the exchange name “International Monetary Market” rather than the Chicago Mercantile Exchange. Matching both names extends the positioning history back to 1986, which gives enough history for the 36-month z-score at the start of the strategy sample.

**Crypto.** The crypto universe contains twelve large coins: Bitcoin (BTC), Ethereum (ETH), BNB, XRP, Cardano (ADA), Dogecoin (DOGE), Solana (SOL), Litecoin (LTC), Chainlink (LINK), Avalanche (AVAX), Terra (LUNA), and FTX Token (FTT). LUNA and FTT, which collapsed in 2022, are deliberately kept to avoid survivorship bias. Funding is positive on most days: average annualised funding is 11.8% for BTC, 13.9% for ETH, and 14.7% for XRP, and funding is positive on 81–88% of days for the main coins (Figure 1). USDC is used only as a sanity check (its average funding is 0.04% a year) and is not traded.



**Figure 1:** Average funding rate by coin

The figure shows the average annualised funding rate of each coin’s USDT-margined perpetual on Binance and, in parentheses, the share of days with positive funding, January 2020 to August 2026. Positive funding means long positions pay short positions. Terra (LUNA) has the highest average funding, and FTX Token (FTT) is the only coin with clearly negative average funding.

## 4 Methodology

### 4.1 Strategy A: crowded G10 carry

Let  $S_{k,t}$  be the spot rate of currency  $k$  in US dollars per unit and  $i_{k,t}$  its three-month interest rate in percent per year. The monthly excess return of a long position in currency  $k$  is

$$rx_{k,t} = \left(1 + \frac{i_{k,t-1}}{1200}\right) \frac{S_{k,t}}{S_{k,t-1}} - \left(1 + \frac{i_{US,t-1}}{1200}\right), \quad (1)$$

which is the return on a one-month forward position under covered interest parity.

At each month-end  $t$  we rank the seven currencies by the interest differential  $i_{k,t} - i_{US,t}$  and assign weight  $w_{k,t} = +0.5$  to each of the two highest and  $w_{k,t} = -0.5$  to each of the two lowest. The portfolio has no net dollar exposure, and its return in month  $t + 1$  is  $r_{t+1} = \sum_k w_{k,t} rx_{k,t+1}$ .

**Crowding signal.** For each currency we compute speculators' net position scaled by open interest from the last CFTC report available at month-end,

$$pos_{k,t} = \frac{\text{non-commercial long}_{k,t} - \text{non-commercial short}_{k,t}}{\text{open interest}_{k,t}}, \quad (2)$$

and standardise it over a rolling 36-month window (minimum 24 months):  $z_{k,t} = (pos_{k,t} - \bar{pos}_k) / \sigma_k$ . The CFTC report records Tuesday positions and is published on Friday, so we use each report only from its release date. Portfolio crowding is

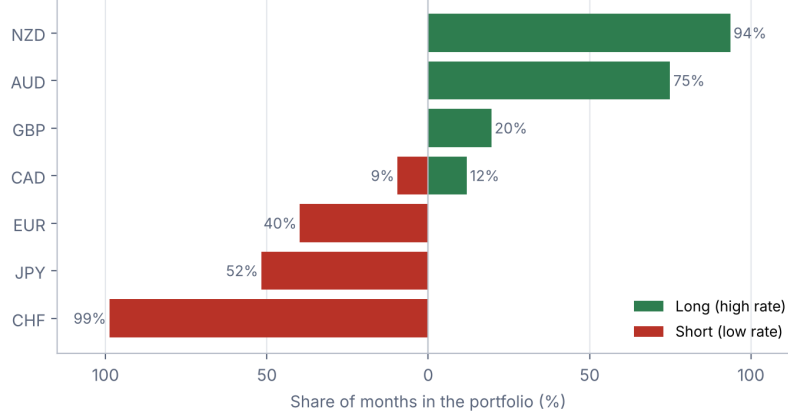
$$\text{crowd}_t = \sum_k w_{k,t} z_{k,t}, \quad (3)$$

the average z-score of the currencies we buy minus the average z-score of those we sell. A high value means speculators are already long our longs and short our shorts. The crowd-filtered strategy multiplies all weights by 0.5 when  $\text{crowd}_t > 1$ .

Figure 2 shows how stable the portfolio is. In practice it is long the New Zealand dollar in 94% of months and the Australian dollar in 75%, and short the Swiss franc in 99% of months, often together with the yen (52%) or the euro (40%). The Canadian dollar and the pound move between the legs as their rates change relative to the others. Thirty-three of the 328 months are classified as crowded.

### 4.2 Strategy B: crypto funding carry

Let  $f_{c,t}$  be the average daily funding rate of coin  $c$  over the seven days up to day  $t - 1$ . A coin is tradable on a day only if it has a price and positive volume, which removes LUNA after May 2022 and FTT after November 14, 2022. Every Sunday we assign weight  $-0.5/k$  to the  $k$  coins with the



**Figure 2:** Strategy A: how often each currency is held

The figure shows the share of the 327 holding months, May 1999 to August 2026, in which each currency is in the long leg (two highest interest rates) or the short leg (two lowest interest rates) of Strategy A.

highest  $f_{c,t}$  and  $+0.5/k$  to the  $k$  coins with the lowest, where  $k$  is one third of the tradable coins. Each leg usually holds three coins (2,151 of 2,403 days); it holds two in early 2020 and briefly four in 2022. Positions are held for seven days, with daily profit and loss

$$\text{pnl}_{c,t+1} = w_{c,t} \frac{P_{c,t+1} - P_{c,t}}{P_{c,t}} - w_{c,t} \text{funding}_{c,t+1}. \quad (4)$$

A short position ( $w < 0$ ) therefore receives funding when funding is positive. We decompose net returns into a funding component, a price component, and costs. There is no separate crowding filter in Strategy B, because the funding rate is itself the crowding signal.

### 4.3 Transaction costs and performance measures

Costs are proportional to turnover,  $\text{cost}_t = c \sum_k |w_{k,t} - w_{k,t-1}|$ , with  $c = 3$  basis points for G10 currencies and  $c = 5$  basis points for crypto perpetuals. We report annualised mean, volatility, and Sharpe ratio (scaled by 12 for monthly and 365 for daily data), maximum drawdown, skewness, worst month or week, hit rate, and annual turnover. Because the strategies are dollar-neutral and earn excess returns, the Sharpe ratio is the mean divided by volatility.

### 4.4 Risk regressions

We estimate the following regressions with Newey and West (1987) standard errors (3 lags for monthly data, 7 lags for daily data):

$$r_t^A = \alpha + \beta_{RX} RX_t + \beta_{vol} \Delta Vol_t + \varepsilon_t, \quad \text{monthly, 1999:05–2021:05,} \quad (5)$$

$$r_t^B = \alpha + \beta_{BTC} r_t^{BTC} + \varepsilon_t, \quad \text{daily, 2020–2026,} \quad (6)$$

where  $RX_t$  is the dollar factor and  $\Delta Vol_t$  the equity-volatility innovation from [Lustig et al. \(2011\)](#). The factor data end in May 2021, which limits the sample of the FX regression.

## 4.5 Combined portfolio

To compare the strategies on equal terms, we scale each to 10% annual volatility using trailing volatility known at  $t - 1$  (36 months for A, 12 months for B) and combine them 50/50.

## 4.6 Keeping the tests honest

Several features of the design limit data mining. All parameters (number of currencies per leg, z-score window, crowding threshold, exposure cut, funding lookback, rebalancing frequency, and costs) were fixed before the backtests and were not changed afterwards. Results are split into an in-sample period (1999–2010 for A, 2020–2023 for B) and an out-of-sample period (2011–2026 for A, 2024–2026 for B). Weights set at the end of period  $t$  earn the return of  $t + 1$ ; CFTC data are used only after their release; and crypto signals use funding up to the previous day. Robustness grids are reported in full but were not used to choose the headline parameters.

# 5 Results: Foreign Exchange Carry

## 5.1 Carry earns a premium and replicates the carry factor (H-A1)

Table [IV](#) reports the performance of Strategy A. Over the full sample, plain carry earns 3.27% a year net of costs with 8.91% volatility, a Sharpe ratio of 0.37, negative skewness of  $-0.49$ , and a maximum drawdown of  $-37\%$ . Returns are positive in 60% of months. The mean return is marginally significant ( $t = 1.92$  over 327 months), a reminder that even a Sharpe ratio of 0.37 is hard to distinguish from zero in 27 years of data. Performance is stronger in the in-sample period (Sharpe 0.47) than out of sample (0.27), consistent with the broad decline in G10 carry returns after the financial crisis. Figure [3](#) shows the growth of one dollar.

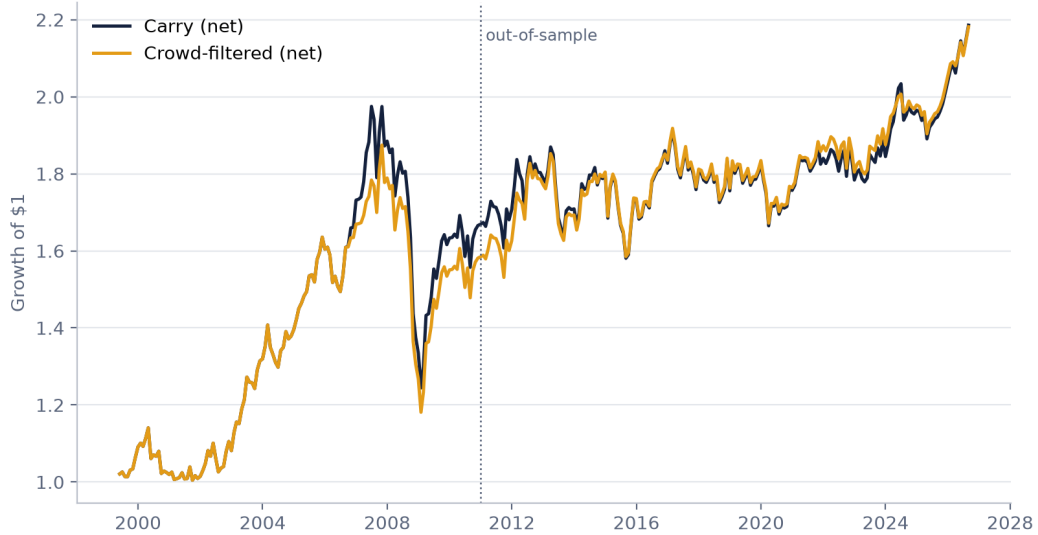
Strategy A closely reproduces the academic carry factor. Over May 1999 to May 2021, its correlation with the [Lustig et al. \(2011\)](#) developed-country HML portfolio is 0.82 (Figure [4](#)). Over this window, Strategy A earns 3.22% a year with a Sharpe ratio of 0.34, against 2.34% and 0.23 for the LRV factor. The evidence supports H-A1.

Figure [5](#) shows where the carry return comes from, currency by currency. Summed over the sample, the gross return is 90 percentage points: interest-rate differentials contribute 96 points and spot moves  $-6$ . The two commodity currencies earn on both counts, with AUD contributing 45 points and NZD 40. The funding side is different. Shorting the Swiss franc collects 25 points of interest but loses 41 points as the franc appreciates, a net loss of 16 points, and the yen’s 18 points

**Table IV:** Strategy A: G10 Carry and the Crowding Filter

| Strategy       | Period                  | Mean (%) | Vol (%) | Sharpe      | Max DD (%) | Skew  | Turnover |
|----------------|-------------------------|----------|---------|-------------|------------|-------|----------|
| Carry          | Full 1999:05–2026:08    | 3.27     | 8.91    | 0.37        | −37.0      | −0.49 | 1.2      |
| Carry          | In-sample 1999–2010     | 4.96     | 10.55   | 0.47        | −37.0      | −0.71 | 1.6      |
| Carry          | Out-of-sample 2011–2026 | 2.01     | 7.46    | 0.27        | −15.5      | −0.10 | 0.9      |
| Crowd-filtered | Full                    | 3.23     | 8.55    | 0.38        | −37.0      | −0.45 | 2.0      |
| Crowd-filtered | In-sample               | 4.47     | 10.12   | 0.44        | −37.0      | −0.69 | 1.8      |
| Crowd-filtered | Out-of-sample           | 2.31     | 7.16    | <b>0.32</b> | −14.4      | −0.02 | 2.1      |

This table reports annualised net-of-cost performance of Strategy A. Carry buys the two G10 currencies with the highest three-month interest rate and sells the two with the lowest, with 50% of capital in each leg. The crowd-filtered strategy halves all positions when the CFTC crowding measure exceeds one. Costs are 3 basis points per unit of turnover. Turnover is annual, in multiples of capital.

**Figure 3:** Strategy A: growth of \$1, carry and crowd-filtered carry (net of costs)

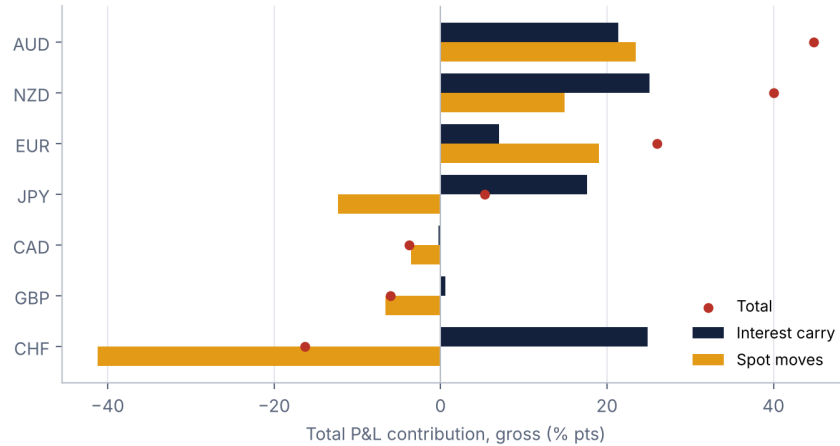
The figure plots the cumulative value of \$1 invested in plain G10 carry and in crowd-filtered carry, net of 3 basis point costs, from May 1999 to August 2026. The dotted line marks the start of the out-of-sample period in January 2011.



**Figure 4:** Strategy A and the Lustig–Roussanov–Verdelhan carry factor

The figure compares the cumulative return of Strategy A (net) with the developed-country high-minus-low carry factor of [Lustig et al. \(2011\)](#), May 1999 to May 2021. The monthly correlation is 0.82.

of interest are mostly offset by its appreciation. This is the classic carry pattern: the premium is the interest differential, and the risk sits in sudden rallies of the low-yield funding currencies.



**Figure 5:** Strategy A: profit and loss by currency

The figure splits the gross profit and loss of Strategy A by currency into the interest-rate differential earned or paid (interest carry) and the spot-rate change (spot moves), summed over monthly returns from May 1999 to August 2026, in percentage points. Dots show the total for each currency.

## 5.2 Does crowding predict weak carry? (H-A2)

Table [V](#) compares carry in the month after crowded and non-crowded readings. After the 294 non-crowded months, carry earns 0.30% in the next month on average. After the 33 crowded months it

earns only 0.02%, with similar volatility and more negative skewness ( $-0.71$  against  $-0.46$ ). The difference is economically meaningful but not statistically significant: a Welch test of equal means gives  $t = -0.57$  ( $p = 0.58$ ). Part of the gap also reflects timing. The 294 non-crowded months include 82 months before mid-2006 in which the crowding measure is unavailable and carry did well (0.51% a month). Comparing only months in which the measure is available, carry earns 0.22% after non-crowded readings against 0.02% after crowded ones ( $t = -0.39$ ).

**Table V:** Next-Month Carry After Crowded and Non-Crowded Months

|                                                                | Months | Mean (%)    | Volatility (%) | Skewness |
|----------------------------------------------------------------|--------|-------------|----------------|----------|
| Not crowded                                                    | 294    | 0.30        | 2.56           | $-0.46$  |
| Crowding measure available                                     | 212    | 0.22        | 2.65           | $-0.41$  |
| Crowding measure missing (1999–2006)                           | 82     | 0.51        | 2.33           | $-0.60$  |
| Crowded                                                        | 33     | <b>0.02</b> | 2.67           | $-0.71$  |
| Difference, crowded minus not crowded: $-0.28$ ( $t = -0.57$ ) |        |             |                |          |

This table reports the monthly mean, volatility, and skewness of plain carry returns in month  $t + 1$ , conditional on whether the crowding measure  $\text{crowd}_t$  exceeded one at the end of month  $t$ . Months in which the crowding measure is missing are counted as not crowded. The  $t$ -statistic is from a Welch test of equal means. Returns are net of costs, May 1999 to August 2026.

As a trading rule, the crowding filter helps modestly. It leaves the full-sample Sharpe ratio almost unchanged (0.37 to 0.38) but raises the out-of-sample Sharpe ratio from 0.27 to 0.32 and slightly reduces volatility and negative skewness (Table IV). The gain is not statistically significant: the monthly return difference between the two strategies out of sample averages 0.30% a year with  $t = 1.00$ . Table VI shows that this improvement is not an artefact of the chosen parameters: every combination of threshold (0.5, 1.0, 1.5) and exposure cut (to 0% or 50%) beats plain carry out of sample, with Sharpe ratios between 0.30 and 0.39.

**Table VI:** Robustness of the Crowding Filter

| Crowding threshold      | Exposure when crowded      | Sharpe, full sample | Sharpe, out-of-sample |
|-------------------------|----------------------------|---------------------|-----------------------|
| 0.5                     | 0%                         | 0.37                | 0.31                  |
| 0.5                     | 50%                        | 0.38                | 0.30                  |
| 1.0                     | 0%                         | 0.38                | 0.37                  |
| <b>1.0</b>              | <b>50% (pre-specified)</b> | <b>0.38</b>         | <b>0.32</b>           |
| 1.5                     | 0%                         | 0.35                | 0.39                  |
| 1.5                     | 50%                        | 0.36                | 0.33                  |
| Plain carry (no filter) |                            | 0.37                | 0.27                  |

This table reports net Sharpe ratios of the crowd-filtered strategy for different crowding thresholds and exposure levels. The pre-specified setting is in bold. The grid is reported for robustness and was not used to choose the headline parameters.

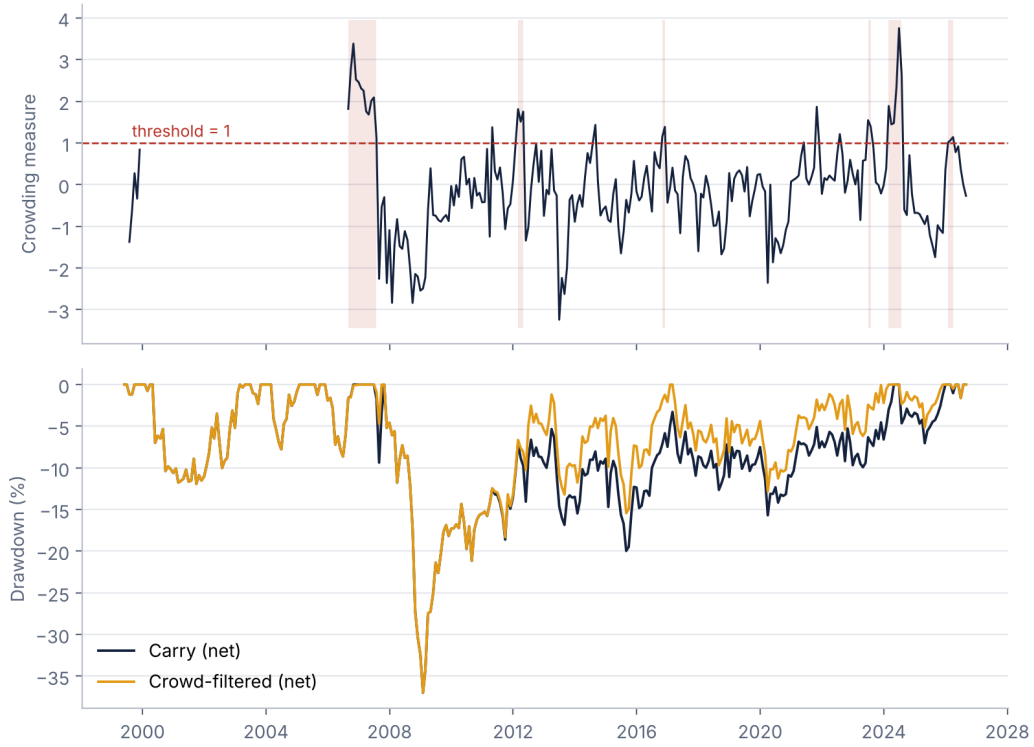
The filter does not, however, avoid the largest crash. The maximum drawdown is  $-37\%$  with or without it, because the 2008 unwind began from a non-crowded reading. Figure 6 shows why. Crowded readings cluster in 2006–2007 and 2024–2026, with a few isolated months in between. Throughout 2008 the measure was negative (between  $-0.8$  and  $-2.8$ ): speculators were positioned

against the carry trade when it crashed, so the filter never cut exposure and the two drawdown paths are identical in 2008. The figure also shows a data gap: because the CFTC series for the New Zealand dollar is missing in 2000–2003 and the z-score needs 24 months of history, the crowding measure is unavailable from June 1999 to July 2006 (81 months). In that period the filter is inactive and the crowd-filtered strategy equals plain carry, so all 33 crowded months fall after mid-2006. Because a binary split with only 33 crowded months has little power, Table VII reports more powerful tests that use all of the information in the crowding measure. Panel A regresses each currency’s next-month excess return on its own crowding z-score, using all 2,149 currency-months with data. The coefficient is small and insignificant whether we pool, add currency fixed effects, or add month fixed effects that compare currencies within the same month. When crowding is signed in the direction of carry, so that a positive value means speculators are long a high-rate currency or short a low-rate one, the coefficient is again insignificant ( $t = 0.54$ ). Panel B uses the continuous portfolio crowding measure. It does not predict carry returns over one, three, six, or twelve months, and it does not predict the probability of a crash month (the worst 10% of months) or the 10th percentile of the return distribution. If anything, the signs point the other way: crowding into high-rate currencies is followed by slightly *higher* returns, as in a momentum effect. We therefore classify H-A2 as “not supported”: CFTC crowding does not predict weaker carry or carry crashes in our sample.

**Table VII:** Does CFTC Crowding Predict Carry? Formal Tests

| Test                                                                                 | Coefficient | $t$ -statistic | $N$   |
|--------------------------------------------------------------------------------------|-------------|----------------|-------|
| <i>Panel A: currency level, next-month excess return (%) on own crowding z-score</i> |             |                |       |
| Pooled, controlling for the rate differential                                        | 0.043       | 0.45           | 2,149 |
| With currency fixed effects                                                          | 0.045       | 0.46           | 2,149 |
| With month fixed effects                                                             | −0.024      | −0.65          | 2,149 |
| Carry-signed return on carry-signed crowding                                         | 0.050       | 0.54           | 2,149 |
| <i>Panel B: portfolio level, on the crowding measure <math>crowd_t</math></i>        |             |                |       |
| Next-month carry return (%)                                                          | 0.009       | 0.04           | 245   |
| Carry return over the next 3 months (%)                                              | 0.308       | 0.54           | 241   |
| Carry return over the next 6 months (%)                                              | 0.386       | 0.44           | 235   |
| Carry return over the next 12 months (%)                                             | 0.074       | 0.06           | 223   |
| Probability of a crash month (worst 10%), linear                                     | −0.025      | −0.85          | 245   |
| 10th percentile of next-month return (quantile regression)                           | 0.443       | 1.09           | 245   |

Panel A regresses the next-month excess return of each currency on its own CFTC crowding z-score  $z_{k,t}$  and the interest-rate differential, with standard errors clustered by month (by currency in the month fixed-effects specification). In the carry-signed specification, returns and z-scores are multiplied by the sign of the interest-rate differential. Panel B regresses net returns of Strategy A on the portfolio crowding measure, using only months in which it is available (from August 2006);  $t$ -statistics use Newey–West standard errors with lags equal to the horizon plus two. A crash month is one in the bottom decile of monthly returns. The code is in `paper/extra_tests.py`.



**Figure 6:** Strategy A: the crowding measure and drawdowns

The top panel plots the portfolio crowding measure  $\text{crowd}_t$ ; shaded months are crowded ( $\text{crowd}_t > 1$ ), when the filter halves all positions. The measure is missing from June 1999 to July 2006. The bottom panel plots the drawdown of plain carry and crowd-filtered carry, net of costs, May 1999 to August 2026.

### 5.3 When does FX carry lose money? (H-A3)

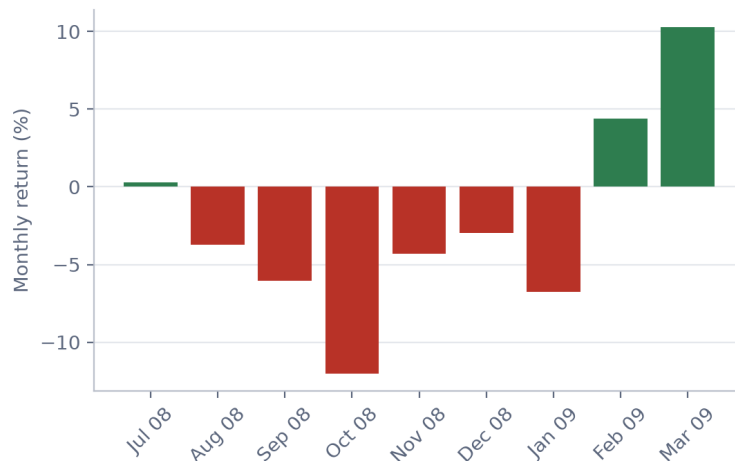
Table VIII regresses monthly net carry returns on the dollar factor and on equity-volatility innovations. Carry loads positively on the dollar factor (0.35,  $t = 4.1$ ) and negatively on volatility shocks ( $-1.55$ ,  $t = -3.7$ ). Together the two factors explain 18% of the variation in carry returns, and the monthly alpha of 0.25% is not statistically significant ( $t = 1.7$ ). The evidence supports H-A3 and the view of Menkhoff et al. (2012) that carry is compensation for exposure to global volatility risk rather than a free lunch.

**Table VIII:** Strategy A: Exposure to Dollar and Volatility Risk

| Model           | Dollar factor (RX) | Volatility innovation | Monthly alpha (%) | $R^2$ |
|-----------------|--------------------|-----------------------|-------------------|-------|
| RX only         | 0.42<br>(4.4)      |                       | 0.26<br>(1.6)     | 0.12  |
| Volatility only |                    | $-1.93$<br>( $-4.3$ ) | 0.26<br>(1.6)     | 0.10  |
| Both            | 0.35<br>(4.1)      | $-1.55$<br>( $-3.7$ ) | 0.25<br>(1.7)     | 0.18  |

This table reports OLS regressions of monthly net returns of Strategy A on the dollar factor RX and the equity-volatility innovation of Lustig et al. (2011). Newey–West  $t$ -statistics with three lags are in parentheses. The sample is May 1999 to May 2021 (264 months), limited by the availability of the factor data.

Figure 7 shows the 2008 episode. From July 2008 to March 2009 the strategy lost 20.5% as the funding currencies, the yen and the Swiss franc, rallied against the Australian and New Zealand dollars. The worst single months were October 2008 ( $-12.0\%$ ), August 2007 ( $-7.8\%$ ), May 2000 ( $-7.0\%$ ), January 2009 ( $-6.8\%$ ), March 2008 ( $-6.5\%$ ), and January 2015 ( $-6.4\%$ , the removal of the Swiss franc floor). All are periods of rising global risk aversion or sudden moves in a funding currency.



**Figure 7:** The 2008 carry crash

The figure shows monthly net returns of Strategy A from July 2008 to March 2009, during which the strategy lost 20.5%.

## 6 Results: Crypto Funding Carry

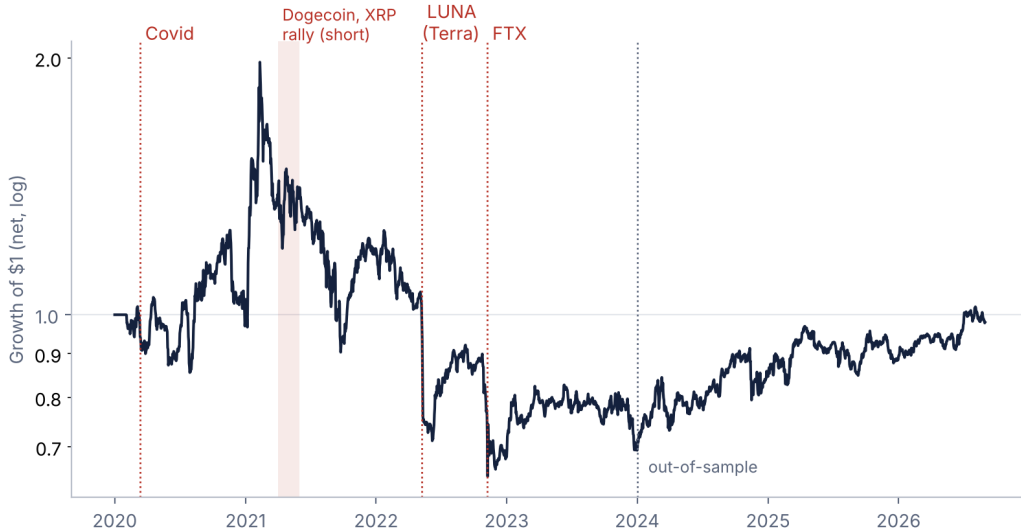
### 6.1 Performance and the source of returns (H-B1)

Table IX reports the performance of Strategy B. Before costs it earns 6.38% a year with 29.9% volatility (Sharpe 0.21). Turnover is high, about 45 times capital a year, so costs of 5 basis points take 2.25 percentage points a year, leaving 4.13% net and a Sharpe ratio of 0.14. The maximum drawdown is  $-67\%$ . The mean return is not statistically different from zero ( $t = 0.35$  over 2,402 days, and  $t = 1.32$  out of sample). Performance differs sharply between subperiods: the strategy loses money in 2020–2023 (Sharpe  $-0.07$ ) and earns 13.9% a year with a Sharpe ratio of 0.81 in 2024–2026. Figure 8 plots the growth of one dollar with the main events.

**Table IX:** Strategy B: Crypto Funding Carry

|       | Period                  | Mean (%) | Vol (%) | Sharpe      | Max DD (%) | Worst week (%) | Turnover |
|-------|-------------------------|----------|---------|-------------|------------|----------------|----------|
| Gross | Full 2020:02–2026:08    | 6.38     | 29.87   | 0.21        | −66.0      | −27.2          | 45       |
| Net   | Full                    | 4.13     | 29.87   | <b>0.14</b> | −67.3      | −27.3          | 45       |
| Net   | In-sample 2020–2023     | −2.52    | 36.05   | −0.07       | −67.3      | −27.3          | 45       |
| Net   | Out-of-sample 2024–2026 | 13.87    | 17.21   | <b>0.81</b> | −13.3      | −9.2           | 44       |

This table reports annualised performance of Strategy B, which every Sunday shorts the third of coins with the highest seven-day average funding rate and buys the third with the lowest, with 50% of capital in each leg, and holds for seven days. Net returns deduct 5 basis points per unit of turnover. Statistics use daily returns annualised with 365 days. Turnover is annual, in multiples of capital.



**Figure 8:** Strategy B: growth of \$1 (net of costs, log scale)

The figure plots the cumulative value of \$1 invested in Strategy B, net of 5 basis point costs, February 2020 to August 2026. The shaded band marks the Dogecoin and XRP rally of April–May 2021, when the strategy was short both coins. The main collapses (Terra in May 2022 and FTX in November 2022) are marked.

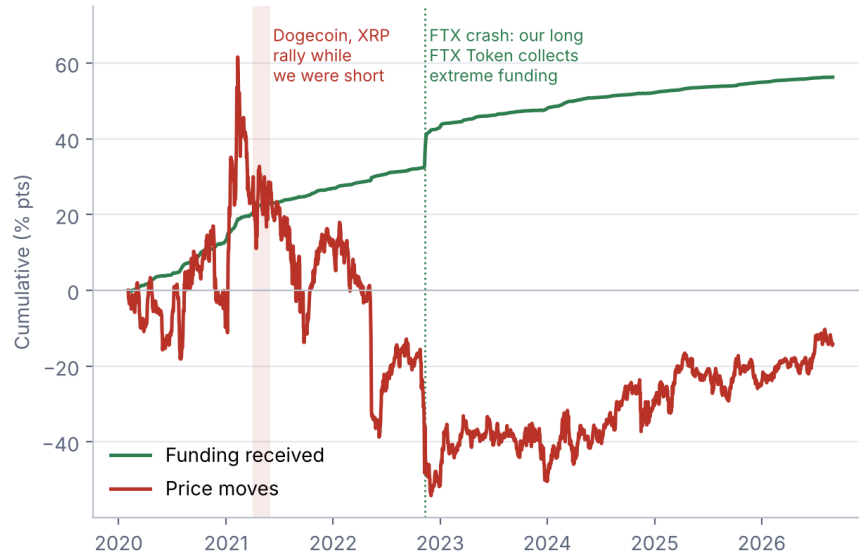
Table X decomposes the return. Over the full sample, the strategy collects 8.55% a year in

funding, as H-B1 predicts, but loses 2.17% a year on price moves and 2.25% on costs. In 2020–2023 the funding income (12.3%) was entirely offset by price losses (−12.5%), as the coins the strategy shorted for their high funding, such as Dogecoin and XRP, kept rallying. Since 2024 funding has been lower (3.1%) and most of the gain came from prices (13.0%). Figure 9 shows the two components over time. Figure 10 brings these results together: the gap between gross and net value is the cost of weekly rebalancing, the funding component rises steadily (with a jump at the FTX collapse), and the price component drives both the drawdown and the recovery. We classify H-B1 as “mixed”: the strategy collects real funding, but price moves decide the result.

**Table X:** Strategy B: Where the Return Comes From

| Period                  | Funding (%) | Price (%) | Costs (%) | Net (%) |
|-------------------------|-------------|-----------|-----------|---------|
| Full 2020–2026          | 8.55        | −2.17     | −2.25     | 4.13    |
| In-sample 2020–2023     | 12.26       | −12.51    | −2.27     | −2.52   |
| Out-of-sample 2024–2026 | 3.10        | 12.99     | −2.22     | 13.87   |

This table decomposes the annualised net return of Strategy B into funding received or paid, price returns, and transaction costs, in percent per year.



**Figure 9:** Strategy B: cumulative funding received and price returns

The figure plots the cumulative funding and price components of Strategy B. The dotted line marks the FTX collapse in November 2022, when the strategy was long FTX Token and collected extreme funding.

Table XI shows the robustness of Strategy B to the funding lookback and rebalancing frequency. All six settings have a positive out-of-sample Sharpe ratio (0.41 to 1.29), and the pre-specified seven-day weekly setting lies in the bottom half of the full-sample results. Shorter lookbacks and daily rebalancing perform better even after costs, which suggests that the funding signal decays quickly.



**Figure 10:** Strategy B: gross and net value, drawdown, and return components

The top panel plots the growth of \$1 in Strategy B before and after 5 basis point costs (log scale); the dashed line marks the start of the out-of-sample period in January 2024. The middle panel plots the net drawdown from the previous peak. The bottom panel plots the cumulative sum of the daily funding and price components, in percent. February 2020 to August 2026.

**Table XI:** Strategy B: Robustness to Lookback and Rebalancing

| Funding lookback | Rebalancing                   | Sharpe, full sample | Sharpe, out-of-sample |
|------------------|-------------------------------|---------------------|-----------------------|
| 3 days           | Daily                         | 0.68                | 0.41                  |
| 3 days           | Weekly                        | 0.49                | 1.29                  |
| 7 days           | Daily                         | 0.56                | 0.92                  |
| <b>7 days</b>    | <b>Weekly (pre-specified)</b> | <b>0.14</b>         | <b>0.81</b>           |
| 30 days          | Daily                         | 0.28                | 0.83                  |
| 30 days          | Weekly                        | 0.07                | 0.42                  |

This table reports net Sharpe ratios of Strategy B for different funding lookback windows and rebalancing frequencies. The pre-specified setting is in bold. The grid was not used to choose the headline parameters.

## 6.2 What drives crypto carry losses? (H-B2)

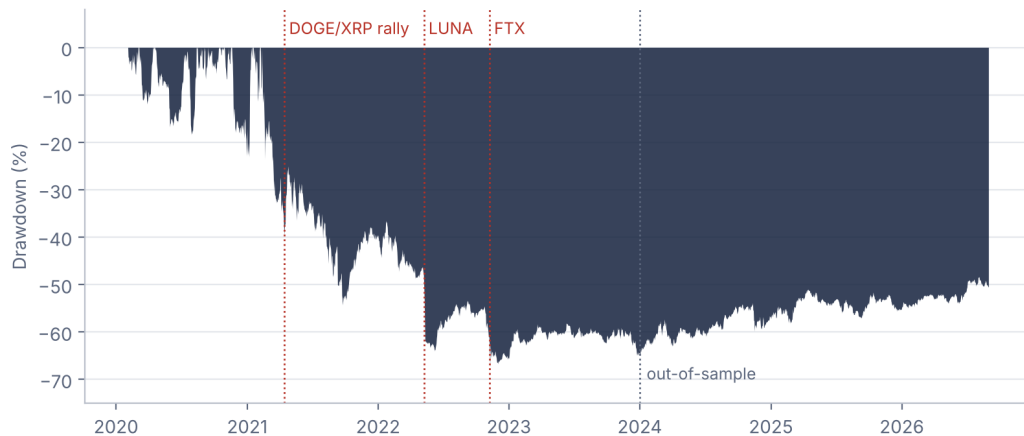
Figure 11 plots the drawdown of Strategy B. The drawdown builds up in 2021 and 2022, during the Dogecoin and XRP rally and the Terra and FTX collapses, and reaches  $-67\%$  on November 9, 2022. The strategy has not regained its 2021 peak: despite the gains since 2024, it ends August 2026 about 55% below it. Crypto carry has no directional market exposure. A daily regression on Bitcoin returns gives a beta of  $-0.02$  (Newey–West  $t = -0.75$ ) and an  $R^2$  of 0.001. Because the claim here is that the beta is zero, the relevant statistic is the width of its confidence interval rather than the  $t$ -statistic. The 95% interval is  $[-0.06, 0.03]$ , which rules out even a small exposure. Table XII shows that the result holds at weekly and monthly horizons, when lagged Bitcoin returns are added to allow for slow price adjustment, and on days when Bitcoin falls. In the 5% of days on which Bitcoin fell most (by 7.2% on average), the strategy earned  $+0.15\%$ , and in Bitcoin’s worst 10% of weeks it lost an insignificant 0.6%. Exposure is not constant, however: the rolling one-year beta ranges from  $-0.21$  to  $+0.19$ , and the size of daily strategy returns is correlated with the size of Bitcoin’s moves (0.20), so the strategy is more volatile in turbulent markets even though it has no directional exposure. Its losses are instead coin-specific. The worst week, ending May 15, 2022, cost 27.3%. In that week the strategy was *long* Terra: traders were shorting LUNA so heavily that its seven-day funding had turned negative, about  $-20\%$  a year, and the rule placed it in the long leg just before it collapsed. The other worst weeks ended September 12, 2021 ( $-12.3\%$ ), November 13, 2022 ( $-11.6\%$ , the FTX collapse), November 29, 2020 ( $-11.0\%$ ), and August 15, 2021 ( $-10.0\%$ ).

**Table XII:** Strategy B: Exposure to Bitcoin

| Specification                        | Beta        | $t$ -statistic | 95% interval      |
|--------------------------------------|-------------|----------------|-------------------|
| Daily                                | $-0.018$    | $-0.78$        | $[-0.064, 0.028]$ |
| Daily, sum of current and three lags | $0.014$     |                |                   |
| Weekly                               | $0.033$     | $0.86$         | $[-0.043, 0.109]$ |
| Monthly                              | $0.056$     | $1.27$         |                   |
| Daily, downside (Bitcoin falling)    | $0.000$     | $0.01$         |                   |
| Daily, upside (Bitcoin rising)       | $-0.036$    | $-1.14$        |                   |
|                                      | Bitcoin (%) | Strategy (%)   | $t$ -statistic    |
| Bitcoin’s worst 5% of days (121)     | $-7.17$     | $0.15$         | $0.74$            |
| Bitcoin’s worst 10% of weeks (35)    | $-13.5$     | $-0.62$        | $-0.75$           |

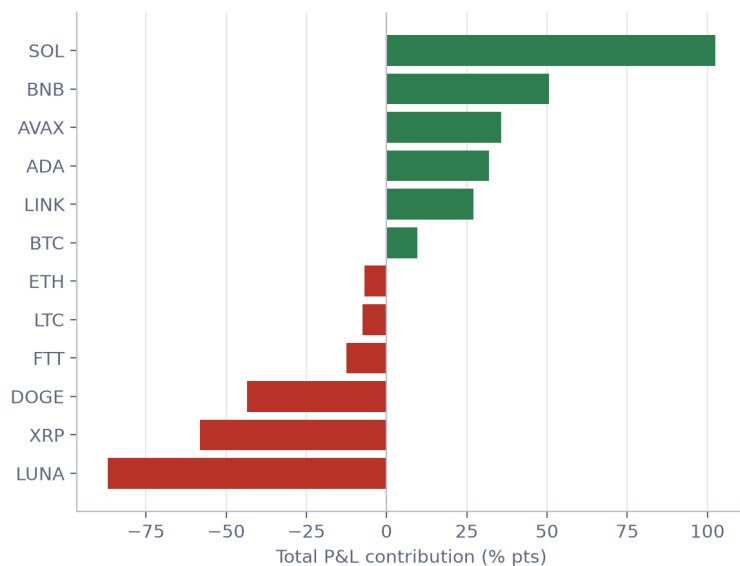
The top panel reports regressions of net returns of Strategy B on Bitcoin spot returns, February 2020 to August 2026, with Newey–West standard errors (five lags for daily and three for weekly and monthly data; the  $t$ -statistic of  $-0.75$  in the text uses the lag choice of `code/05_risk_analysis.py`). In the lagged specification, none of the individual coefficients is significant ( $|t| < 1.1$ ). Downside and upside betas split Bitcoin returns at zero. The bottom panel reports average returns on the days and weeks with the worst Bitcoin returns. The code is in `paper/extra_tests.py`.

Figure 12 shows the total contribution of each coin to the strategy’s profit and loss. Four coins account for most of the losses: Terra ( $-87$  percentage points), XRP ( $-58$ ), Dogecoin ( $-44$ ), and FTX Token ( $-13$ ). The winners are Solana ( $+102$ ), BNB ( $+51$ ), Avalanche ( $+36$ ), Cardano ( $+32$ ), and Chainlink ( $+27$ ). In every case the price component dominates the funding component.



**Figure 11:** Strategy B: drawdown from the previous peak

The figure plots the percentage decline of Strategy B (net of costs) from its previous peak, February 2020 to August 2026. Dotted red lines mark the Dogecoin and XRP rally (April 2021), the Terra collapse (May 2022), and the FTX collapse (November 2022). The grey line marks the start of the out-of-sample period in January 2024.



**Figure 12:** Strategy B: total profit and loss by coin

The figure shows the cumulative contribution of each coin to the net profit and loss of Strategy B, in percentage points, February 2020 to August 2026.

This pattern is the mirror image of FX carry. In FX, carry crashes when a crowded long position in high-yield currencies unwinds. In crypto, the dangerous positions are those in which the strategy takes the other side of a crowd: it is short coins that leveraged traders are crowding to buy (Dogecoin and XRP in 2021), and long coins they are crowding to sell (Terra and FTX Token in 2022). The evidence supports H-B2: crypto carry is market-neutral, but its crashes come from single coins with extreme funding.

## 7 Diversification: Combining FX and Crypto Carry

Table XIII reports the two strategies scaled to 10% volatility and their 50/50 combination over February 2021 to August 2026, the period in which both scaled series are available. The correlation between the scaled strategies is 0.11 (0.12 for unscaled monthly net returns over 78 months). The combined portfolio has volatility of 7.0%, below the 9.3–9.4% of either part, and a maximum drawdown of  $-10.6\%$ , smaller than that of either strategy. Because crypto carry was weak over this period, the combination does not beat FX carry alone on Sharpe ratio (0.51 against 0.71). Figure 13 plots the three series. The low correlation is not an average of offsetting regimes: Figure 14 shows that the 24-month rolling correlation stays between  $-0.24$  and  $+0.20$  throughout the sample. The evidence supports H-D.

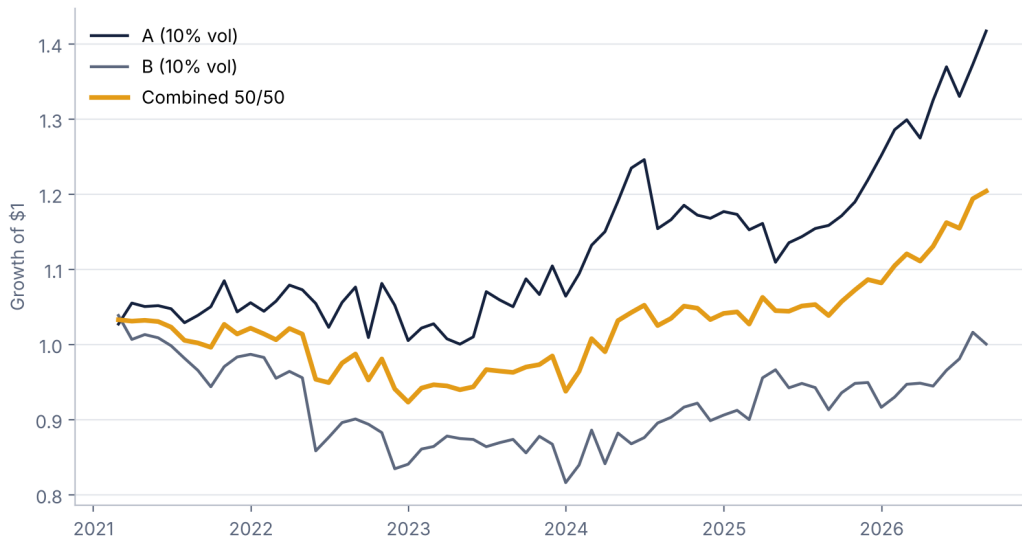
**Table XIII:** Combined Portfolio, 2021:02–2026:08

|                | Mean (%) | Vol (%) | Sharpe | Max DD (%) | Worst month (%) |
|----------------|----------|---------|--------|------------|-----------------|
| A (10% vol)    | 6.70     | 9.43    | 0.71   | $-11.0$    | $-7.4$          |
| B (10% vol)    | 0.44     | 9.28    | 0.05   | $-21.4$    | $-10.2$         |
| Combined 50/50 | 3.57     | 6.96    | 0.51   | $-10.6$    | $-5.9$          |

Each strategy is scaled to 10% annual volatility using trailing volatility known at  $t - 1$  (36 months for A, 12 months for B) and the two are combined with equal weights. Statistics are annualised from 67 monthly net returns.

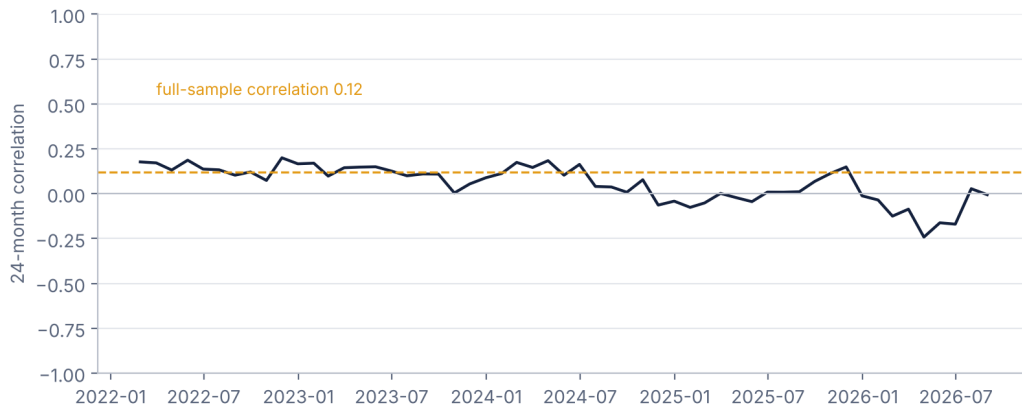
## 8 Robustness

We test whether the main results depend on parameter choices made before the backtests were run. Figure 15 collects the two parameter grids of Tables VI and XI. For FX carry (Panel A), all six combinations of crowding threshold and exposure cut have a full-sample Sharpe ratio between 0.35 and 0.38, close to the 0.37 of plain carry, and every one of them beats plain carry out of sample (0.30 to 0.39 against 0.27). The benefit of the filter is small but does not depend on the exact setting. For crypto carry (Panel B), every setting earns a positive Sharpe ratio both over the full sample and out of sample. The ranking across settings, however, is unstable: the 3-day daily strategy is best over the full sample (0.68) but worst out of sample (0.41), while the 3-day weekly strategy is the reverse. The pre-specified 7-day weekly setting is among the weakest over the full



**Figure 13:** FX carry, crypto carry, and the 50/50 combination

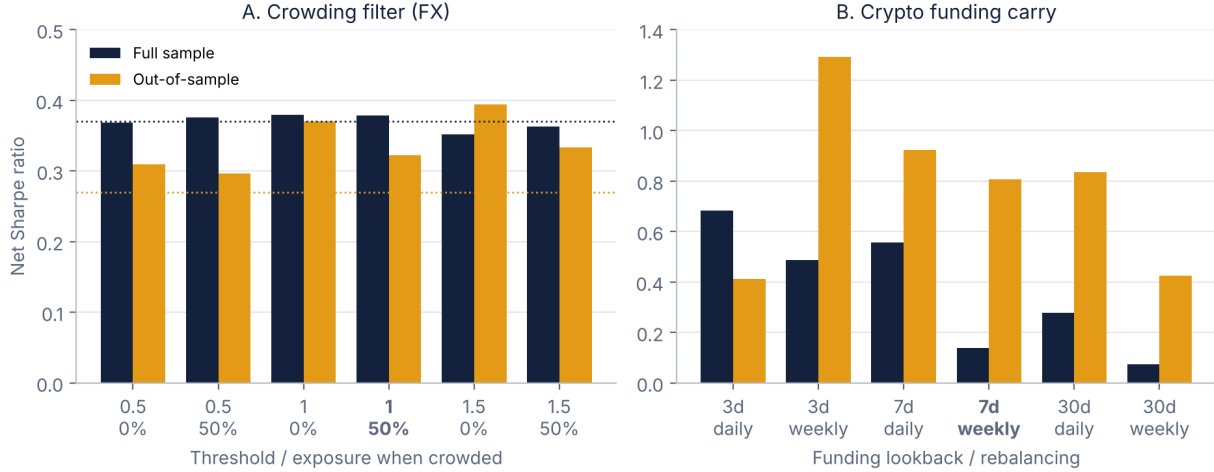
The figure plots the cumulative value of \$1 invested in Strategy A and Strategy B, each scaled to 10% volatility, and in their 50/50 combination, February 2021 to August 2026.



**Figure 14:** Rolling correlation between FX carry and crypto carry

The figure plots the 24-month rolling correlation between monthly net returns of Strategy A and Strategy B. The dashed line is the full-sample correlation of 0.12 over 78 months, March 2020 to August 2026.

sample, so our headline crypto results are, if anything, conservative.



**Figure 15:** Robustness of both strategies to parameter choices

Panel A reports net Sharpe ratios of crowd-filtered FX carry for each crowding threshold and exposure when crowded; the dotted lines mark plain carry (0.37 full sample, 0.27 out-of-sample from 2011). Panel B reports net Sharpe ratios of crypto funding carry for each funding lookback and rebalancing frequency (out-of-sample from 2024). The pre-specified settings are in bold.

We also test one pre-specified risk control for Strategy B, motivated by the coin-specific crashes of Section 6. It has two rules: no coin may exceed one sixth of its leg, with the excess held in cash, and a coin is not eligible for the long leg if its seven-day funding is below  $-50\%$  a year, a sign of crowded shorts in distress. Table XIV shows that this control does not help. It halves volatility and reduces the maximum drawdown from  $-67\%$  to  $-43\%$ , but the net return turns negative ( $-2.2\%$  a year, Sharpe  $-0.16$ ), because the cap halves exposure and the distress rule kept BNB and Solana, the two largest winners, out of the long leg. It also failed in the episode it was designed for: Terra’s funding crossed  $-50\%$  only on May 10, 2022, in the middle of the week in which it collapsed and after the portfolio had been formed.

**Table XIV:** Strategy B With a Pre-Specified Risk Control

|                 | Period         | Mean (%) | Vol (%) | Sharpe | Max DD (%) | Worst week (%) |
|-----------------|----------------|----------|---------|--------|------------|----------------|
| Base            | Full 2020–2026 | 4.13     | 29.87   | 0.14   | −67.3      | −27.3          |
| Base            | OOS 2024–2026  | 13.87    | 17.21   | 0.81   | −13.3      | −9.2           |
| Risk-controlled | Full 2020–2026 | −2.23    | 14.11   | −0.16  | −43.3      | −18.9          |
| Risk-controlled | OOS 2024–2026  | 6.36     | 8.57    | 0.74   | −6.9       | −4.6           |

This table compares Strategy B with a version that caps each coin at one sixth of its leg (excess held in cash) and excludes coins with seven-day funding below  $-50\%$  a year from the long leg. OOS is out-of-sample. Both rules were fixed before the test. Returns are net of 5 basis point costs and annualised.

## 9 Discussion

Table [XV](#) summarises the evidence on each hypothesis.

**Table XV:** Summary of Hypotheses and Evidence

| Hypothesis                                   | Verdict       | Evidence                                                                                  |
|----------------------------------------------|---------------|-------------------------------------------------------------------------------------------|
| H-A1: G10 carry earns a premium              | Supported     | Net Sharpe 0.37; correlation 0.82 with the LRV factor                                     |
| H-A2: crowding predicts weaker carry         | Not supported | 0.02% vs 0.30% ( $t = -0.57$ ); panel, horizon, and tail tests insignificant; missed 2008 |
| H-A3: carry loses in volatility spikes       | Supported     | Volatility beta $-1.55$ ( $t = -3.7$ ); alpha not significant                             |
| H-B1: funding carry earns a premium          | Mixed         | Funding $+8.5\%$ a year, but net Sharpe 0.14; price moves dominate                        |
| H-B2: no BTC exposure; coin-specific crashes | Supported     | BTC beta $-0.02$ , 95% interval $[-0.06, 0.03]$ ; Terra $-87$ points                      |
| H-D: FX and crypto carry diverge             | Supported     | Correlation 0.11; combined volatility 7.0% vs about 9%                                    |

**Interpretation.** In foreign exchange, our results agree with the risk-based view of carry. G10 carry earns a modest premium, tracks the academic carry factor, and loses when volatility rises. Crowding measured from CFTC positions, however, does not predict carry returns or crashes. The point estimates go in the expected direction in the simplest comparison, but none of our tests is significant, and the largest crash in our sample started from a position that did not look crowded. The 2008 unwind was driven by a global funding shock that hit carry whether or not speculators were heavily positioned, consistent with [Brunnermeier and Pedersen \(2009\)](#). Our null result should be read alongside studies that do find a role for crowding. [Sokolovski \(2016\)](#) and [Pojarliev and Levich \(2011\)](#) measure crowding from currency co-movement and fund returns, [Cheema-Fox et al. \(2025\)](#) from institutional flows, and [Kutuk and van Wijnbergen \(2025\)](#) price crash risk from options. These measures are faster and capture investors that the CFTC futures report misses, such as banks’ forward books, real-money investors, and Japanese retail margin traders. Individual episodes show both sides. Our crowding measure was high before the August 2007 loss ( $-7.8\%$ ) and before the July 2024 loss ( $-4.7\%$ ) that began the yen unwind documented by [Aquilina et al. \(2024\)](#), and in both cases the filter halved exposure. But high readings also preceded many ordinary months, and the 2008 crash began from a negative reading. Positioning in futures may therefore identify fragile episodes after the fact without being a reliable systematic signal of when crowding will unwind.

In crypto, the funding rate is a direct and fast measure of crowded leveraged demand. Sorting on it collects real income, but it also places the strategy against the crowd on both sides. Being short crowded longs lost money in the 2021 retail rally, and being long crowded shorts lost money in the 2022 collapses. This is consistent with [Schmeling et al. \(2025\)](#), who find that high funding reflects leveraged retail demand meeting scarce arbitrage capital, and with [Christin et al. \(2023\)](#),

who find that the funding trade suffered most in the Terra and FTX collapses. Our cross-sectional design adds that the losses come from the coins at the extremes of the funding distribution. The fall in funding income after 2024 (from 12.3% to 3.1% a year) matches the compression of the Bitcoin funding premium documented by [Borri et al. \(2025\)](#). Crowding therefore tells an investor *where* the risk sits rather than *when* it will materialise.

**Implications.** For FX investors, CFTC positioning data are not a reliable timing signal; the small out-of-sample gain from the filter is not significant, and protection against volatility shocks matters more. For crypto, sorting on funding alone exposes a portfolio to single-coin collapses. A better design would react faster (the robustness grid points towards shorter lookbacks and daily rebalancing), treat extremely negative funding as a warning of crowded shorts rather than an opportunity, and size positions by coin-level risk. In FX, a rule that reduces exposure when either crowding or implied volatility is high may protect against both kinds of crash, and adding emerging-market currencies would raise the number of crowded episodes and test whether crowding matters more where liquidity is thinner. In crypto, extreme negative funding should be tested as a warning signal in a larger universe that includes coins that later collapsed, and a practical design would also account for slippage, funding caps, and the risk of exchange failure. The combined portfolio could add carry in other asset classes ([Koijen et al., 2018](#)), although the premium in the simplest crypto carry trade has compressed since 2024 ([Borri et al., 2025](#)).

**Limitations.** Our results have several limitations. The crypto sample covers about six and a half years and twelve coins, each leg usually holds three coins, and single-coin events dominate. Strategy B reverses between the in-sample period (Sharpe  $-0.07$ ) and the out-of-sample period ( $0.81$ ), and the out-of-sample gains came mostly from price moves, so they are not evidence of a stable premium. In FX, only 33 months were crowded, which limits the power of the simple comparison, although the panel and continuous tests use the full sample, and the crowding measure is missing from mid-1999 to mid-2006, so the filter is tested only on the last twenty years. Data availability limits the FX sample to start in 1999 and the factor regressions to end in 2021. Finally, transaction costs are proportional estimates; we do not model funding caps, borrowing limits, slippage under stress, or exchange counterparty risk, which was material in the FTX collapse.

## 10 Conclusion

We ask whether carry trades crash when they are crowded, using G10 currencies and crypto perpetual futures. Our answer differs across the two markets. In FX, carry crashes are compensation for exposure to volatility risk, and crowding measured from CFTC positions does not predict carry returns or crashes in any of our tests. In crypto, carry also earns a premium and also crashes, but the crashes come from single coins, mostly from being long a coin that traders are crowding to

short, so crowding tells investors where the danger is rather than when it will hit. Because the two carry trades are nearly uncorrelated, combining them reduces volatility and drawdowns.

Our results point to three directions for future work. First, the crowding that drives FX carry crashes may be better captured by faster measures than weekly futures positions, such as option-implied crash risk (Kutuk and van Wijnbergen, 2025), institutional flows (Cheema-Fox et al., 2025), or co-movement among carry currencies (Sokolovski, 2016), possibly combined with volatility. Second, in crypto, open interest, liquidation data, and funding across exchanges could sharpen the funding signal, and the delta-neutral basis trade (Christin et al., 2023; Gornall et al., 2025) offers funding income without coin-specific price risk. Third, coin-level risk limits and estimated weights in the combined portfolio would bring both strategies closer to what an investor could trade.

**Data availability.** All data used in this paper are public: FRED (Federal Reserve H.10 spot rates and OECD interest rates), the CFTC Commitments of Traders reports, the Binance public data archive, and the carry factor portfolios of Lustig et al. (2011). The raw downloads, cleaned data, code, and the replication package are available at <https://github.com/rezazamani2329/currency-crypto-carry>; Appendix A describes how to reproduce every table and figure.

## References

- Brunnermeier, M. K., S. Nagel, and L. H. Pedersen, 2008, Carry trades and currency crashes, *NBER Macroeconomics Annual* 23, 313–347.
- Brunnermeier, M. K., and L. H. Pedersen, 2009, Market liquidity and funding liquidity, *Review of Financial Studies* 22, 2201–2238.
- Aquilina, M., M. Lombardi, A. Schrimpf, and V. Sushko, 2024, The market turbulence and carry trade unwind of August 2024, BIS Bulletin No. 90, Bank for International Settlements.
- Bank for International Settlements, 2024, Hedge fund exposure to the carry trade, *BIS Quarterly Review*, September 2024.
- Borri, N., Y. Liu, A. Tsyvinski, and X. Wu, 2025, Cryptocurrency as an investable asset class: Coming of age, Working paper, arXiv:2510.14435.
- Burnside, C., M. Eichenbaum, I. Kleshchelski, and S. Rebelo, 2011, Do peso problems explain the returns to the carry trade?, *Review of Financial Studies* 24, 853–891.
- Cheema-Fox, A., E. S. Cuipa, and R. Greenwood, 2025, Flight to safety or flight from carry?, Working paper, Harvard University; *Financial Analysts Journal*, 2026.
- Christin, N., B. R. Routledge, K. Soska, and A. Zetlin-Jones, 2023, The crypto carry trade, Working paper, Carnegie Mellon University.
- Daniel, K., R. J. Hodrick, and Z. Lu, 2017, The carry trade: Risks and drawdowns, *Critical Finance Review* 6, 211–262.
- Fama, E. F., 1984, Forward and spot exchange rates, *Journal of Monetary Economics* 14, 319–338.
- Farhi, E., and X. Gabaix, 2016, Rare disasters and exchange rates, *Quarterly Journal of Economics* 131, 1–52.
- Gornall, W., M. Rinaldi, and Y. Xiao, 2025, Perpetual futures and basis risk: Evidence from cryptocurrency, Working paper, University of British Columbia.
- Hansen, L. P., and R. J. Hodrick, 1980, Forward exchange rates as optimal predictors of future spot rates: An econometric analysis, *Journal of Political Economy* 88, 829–853.
- He, S., A. Manela, O. Ross, and V. von Wachter, 2022, Fundamentals of perpetual futures, Working paper, arXiv:2212.06888.
- Jurek, J. W., 2014, Crash-neutral currency carry trades, *Journal of Financial Economics* 113, 325–347.
- Koijen, R. S. J., T. J. Moskowitz, L. H. Pedersen, and E. B. Vrugt, 2018, Carry, *Journal of Financial Economics* 127, 197–225.
- Kutuk, M. M., and S. van Wijnbergen, 2025, Carry trade and currency crash risk, CEPR Discussion Paper No. 20745.
- Liu, Y., and A. Tsyvinski, 2021, Risks and returns of cryptocurrency, *Review of Financial Studies* 34, 2689–2727.

- Liu, Y., A. Tsyvinski, and X. Wu, 2022, Common risk factors in cryptocurrency, *Journal of Finance* 77, 1133–1177.
- Lustig, H., N. Roussanov, and A. Verdelhan, 2011, Common risk factors in currency markets, *Review of Financial Studies* 24, 3731–3777.
- Lustig, H., N. Roussanov, and A. Verdelhan, 2014, Countercyclical currency risk premia, *Journal of Financial Economics* 111, 527–553.
- Lustig, H., and A. Verdelhan, 2007, The cross section of foreign currency risk premia and consumption growth risk, *American Economic Review* 97, 89–117.
- Menkhoff, L., L. Sarno, M. Schmeling, and A. Schrimpf, 2012, Carry trades and global foreign exchange volatility, *Journal of Finance* 67, 681–718.
- Moskowitz, T. J., Y. H. Ooi, and L. H. Pedersen, 2012, Time series momentum, *Journal of Financial Economics* 104, 228–250.
- Newey, W. K., and K. D. West, 1987, A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix, *Econometrica* 55, 703–708.
- Pedersen, L. H., 2009, When everyone runs for the exit, *International Journal of Central Banking* 5, 177–199.
- Pojarliev, M., and R. M. Levich, 2011, Detecting crowded trades in currency funds, *Financial Analysts Journal* 67, 26–39.
- Schmeling, M., A. Schrimpf, and K. Todorov, 2023, Crypto carry, BIS Working Paper No. 1087, Bank for International Settlements.
- Schmeling, M., A. Schrimpf, and K. Todorov, 2025, Crypto carry, CEPR Discussion Paper No. 20719.
- Sokolovski, V., 2016, Crowds, crashes, and the carry trade, Working paper, Stockholm School of Economics.
- Stein, J. C., 2009, Presidential address: Sophisticated investors and market efficiency, *Journal of Finance* 64, 1517–1548.

## A Replication

All code and data needed to reproduce the results are in the replication package and at <https://github.com/rezazamani2329/currency-crypto-carry>. The scripts in `code/` run in order: 01 and 03 download and clean the FX, CFTC, and Binance data; 02 and 04 run the backtests for Strategies A and B; and 05 runs the factor regressions, coin-level analysis, and the combined portfolio. Running `SKIP_DOWNLOAD=1 python run_all.py` reproduces every table and figure from the cleaned data in `data/clean/`. Table XVI maps each table and figure to its source file.

**Table XVI:** Source Files for Tables and Figures

| Table or figure            | Source file                                                      |
|----------------------------|------------------------------------------------------------------|
| Table IV, Figures 3, 6     | output/strategyA/performance_table.csv, monthly_returns.csv      |
| Table III                  | data/clean/fx_spot_monthly_usd_per_fcu.csv, rates_3m_monthly.csv |
| Table V                    | output/strategyA/crowded_vs_not.csv                              |
| Table VI                   | output/strategyA/robustness_grid.csv                             |
| Table VIII                 | output/risk/A_factor_regressions.csv                             |
| Figure 4                   | output/risk/A_vs_LRV_HML.csv                                     |
| Figure 5                   | data/clean/fx_spot_monthly_usd_per_fcu.csv, rates_3m_monthly.csv |
| Figure 7                   | output/risk/A_worst_months.csv                                   |
| Table IX, Figures 8, 11    | output/strategyC/performance_table.csv, daily_returns.csv        |
| Table X, Figure 9          | output/strategyC/return_decomposition.csv                        |
| Table XI                   | output/strategyC/robustness_grid.csv                             |
| Figure 12                  | output/risk/C_contribution_by_coin.csv                           |
| Figure 10                  | output/strategyC/daily_returns.csv                               |
| Figure 15                  | output/strategy{A,C}/robustness_grid.csv                         |
| Table XIV                  | output/risk/C_risk_control.csv                                   |
| Tables VII, XII            | paper/extra_tests.py (prints all statistics)                     |
| Table XIII, Figures 13, 14 | output/risk/combined_portfolio.csv, combined_monthly_returns.csv |
| Figure 2                   | output/strategyA/weights_carry.csv                               |
| Figure 1                   | output/strategyC/coin_summary.csv                                |

Figures 1, 2, 5, 6, 11, 14, 15, 10, and 13 are drawn from these files by `paper/make_paper_figures.py`. In the code and output folders, the crypto strategy (Strategy B in this paper) is labelled C.